

BoardNet

Board Examination *Communication Network*

421 Project

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1. Work Distribution

The six boards were split evenly across the three group members, each pair chosen to keep related routing/service work together (core + isolated board, the two RIPv2 boards, and the two static/recursive-redundancy boards).

Group Member (Name / ID)	Board(s) / Task Assigned	Details of Work
Debopriyo Karmaker (ID: 23301543)	DHK Board + CTG Board	DHK router configuration (core router); Central DNS server (all A-records); main DHCP server pools serving DHK, CTG, RAJ, SYL; DHK WEB & EMAIL servers; CTG router configuration with default route to DHK; CTG WEB & EMAIL servers; VLSM base addressing plan
Talha Bin Zakir (ID: 23301079)	RAJ Board + SYL Board	RAJ router configuration with RIPv2 peering to DHK; RAJ EMAIL server and printer; SYL router configuration with RIPv2 peering to DHK; SYL EMAIL server and printer; verification of central-switch segment (DHK-RAJ-SYL) reachability
Abdullah Bin Islam (ID: 23301565)	KHU Board + BAR Board	KHU router configuration with static routing and floating recursive-redundancy routes; dedicated KHU DHCP server; KHU EMAIL server and printer; BAR router configuration with static routing and floating recursive-redundancy routes; dedicated BAR DHCP server; BAR EMAIL server and printer; redundancy failover testing (link shutdown demo)

2. Network Topology Diagram

Figure 1 shows the full BoardNet topology as implemented in the submitted Packet Tracer (.pkt) file: six board routers (DHK, RAJ, CTG, SYL, BAR, KHU), the central examination-coordination switch joining DHK-RAJ-SYL, the DHK-CTG isolated link, and the DHK-KHU-BAR triangular link set, with each board's own LAN switch, printer, email server and two PCs.

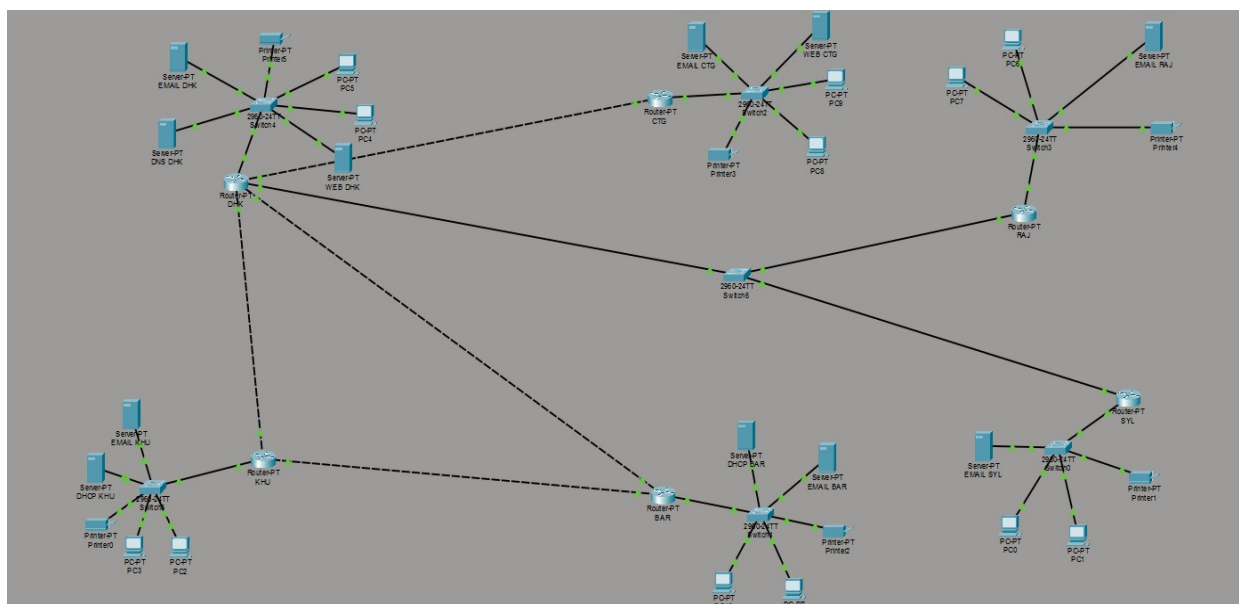


Figure 1. BoardNet full topology (as built in Packet Tracer)

3. Complete VLSM Tree

Base network address: 10.79.0.0/16 (65,536 addresses). Subnets are allocated in descending order of host requirement, each rounded up to the nearest power of two that includes network + broadcast + gateway overhead.

3.1 Allocation Steps

10.79.0.0/16 (65536 addresses) – Base block

```

|
+-- 10.79.0.0/23  -> DHK   (300 hosts needed, 510 usable)   [10.79.0.0 - 10.79.1.255]
|
+-- 10.79.2.0/24  -> CTG   (200 hosts needed, 254 usable)   [10.79.2.0 - 10.79.2.255]
|
+-- 10.79.3.0/24  -> RAJ   (160 hosts needed, 254 usable)   [10.79.3.0 - 10.79.3.255]
|
+-- 10.79.4.0/24  -> SYL   (150 hosts needed, 254 usable)   [10.79.4.0 - 10.79.4.255]
|
+-- 10.79.5.0/25   -> BAR   (120 hosts needed, 126 usable)   [10.79.5.0 - 10.79.5.127]
|
+-- 10.79.5.128/25 -> KHU   (120 hosts needed, 126 usable)   [10.79.5.128 - 10.79.5.255]
|
+-- 10.79.6.0/24   -> Reserved for WAN / inter-router links (further subnetted below)
    |
    +-- 10.79.6.0/29   -> Central switch segment: DHK - RAJ - SYL (6 usable)
    +-- 10.79.6.8/30   -> Point-to-point: DHK - CTG
    +-- 10.79.6.12/30  -> Point-to-point: DHK - KHU
    +-- 10.79.6.16/30  -> Point-to-point: KHU - BAR
    +-- 10.79.6.20/30  -> Point-to-point: DHK - BAR

```

3.2 LAN Subnet Table

Board	Hosts Req.	Hosts+OH	Block Size	Network Address	Subnet Mask	Usable Range	Broadcast
Dhaka (DHK)	300	302	512 (/23)	10.79.0.0	255.255.254.0 (/23)	10.79.0.1 – 10.79.1.254	10.79.1.255
Chattogram (CTG)	200	202	256 (/24)	10.79.2.0	255.255.255.0 (/24)	10.79.2.1 – 10.79.2.254	10.79.2.255
Rajshahi (RAJ)	160	162	256 (/24)	10.79.3.0	255.255.255.0 (/24)	10.79.3.1 – 10.79.3.254	10.79.3.255
Sylhet (SYL)	150	152	256 (/24)	10.79.4.0	255.255.255.0 (/24)	10.79.4.1 – 10.79.4.254	10.79.4.255
Barishal (BAR)	120	122	128 (/25)	10.79.5.0	255.255.255.128 (/25)	10.79.5.1 – 10.79.5.126	10.79.5.127
Khulna (KHU)	120	122	128 (/25)	10.79.5.128	255.255.255.128 (/25)	10.79.5.129 – 10.79.5.254	10.79.5.255

3.3 WAN / Inter-Router Link Subnets

Link	Subnet	Mask	Assigned Addresses	Broadcast
DHK – RAJ – SYL (central switch, multi-access)	10.79.6.0/29	255.255.255.248	DHK .1 · RAJ .2 · SYL .3	10.79.6.7
DHK – CTG (point-to-point)	10.79.6.8/30	255.255.255.252	DHK .9 · CTG .10	10.79.6.11
DHK – KHU (point-to-point)	10.79.6.12/30	255.255.255.252	DHK .13 · KHU .14	10.79.6.15
KHU – BAR (point-to-point)	10.79.6.16/30	255.255.255.252	KHU .17 · BAR .18	10.79.6.19

Link	Subnet	Mask	Assigned Addresses	Broadcast
point)				
DHK – BAR (point-to-point)	10.79.6.20/30	255.255.255.252	DHK .21 · BAR .22	10.79.6.23

4. Full IP Addressing Table

PCs marked "DHCP (client)" receive their address dynamically per the DHCP Configuration in Section 6; all routers, servers, and printers are statically configured.

4.1 Dhaka Board (DHK)

Device	IP Address	Subnet Mask	Default Gateway
Router DHK – Gi6/0 (LAN)	10.79.0.1	255.255.254.0 (/23)	N/A
Router DHK – Fa0/0 (central link)	10.79.6.1	255.255.255.248 (/29)	N/A
Router DHK – Fa1/0 (link to CTG)	10.79.6.9	255.255.255.252 (/30)	N/A
Router DHK – Gi7/0 (link to KHU)	10.79.6.13	255.255.255.252 (/30)	N/A
Router DHK – Gi8/0 (link to BAR)	10.79.6.21	255.255.255.252 (/30)	N/A
Server-PT DNS DHK	10.79.0.2	255.255.254.0	10.79.0.1
Server-PT WEB DHK	10.79.0.3	255.255.254.0	10.79.0.1
Server-PT EMAIL DHK	10.79.0.4	255.255.254.0	10.79.0.1
Printer5	10.79.0.10	255.255.254.0	10.79.0.1
PC4	DHCP (client)	DHCP	DHCP
PC5	DHCP (client)	DHCP	DHCP

4.2 Chattogram Board (CTG)

Device	IP Address	Subnet Mask	Default Gateway
Router CTG – Fa0/0 (link to DHK)	10.79.6.10	255.255.255.252 (/30)	N/A
Router CTG – Gi6/0 (LAN)	10.79.2.1	255.255.255.0 (/24)	N/A
Server-PT WEB CTG	10.79.2.2	255.255.255.0	10.79.2.1
Server-PT EMAIL CTG	10.79.2.3	255.255.255.0	10.79.2.1
Printer3	10.79.2.10	255.255.255.0	10.79.2.1
PC8	DHCP (client, via DHK)	DHCP	DHCP
PC9	DHCP (client, via DHK)	DHCP	DHCP

4.3 Rajshahi Board (RAJ)

Device	IP Address	Subnet Mask	Default Gateway
Router RAJ – Fa0/0 (central link)	10.79.6.2	255.255.255.248 (/29)	N/A
Router RAJ – Gi6/0 (LAN)	10.79.3.1	255.255.255.0 (/24)	N/A
Server-PT EMAIL RAJ	10.79.3.2	255.255.255.0	10.79.3.1
Printer4	10.79.3.10	255.255.255.0	10.79.3.1
PC6	DHCP (client, via DHK)	DHCP	DHCP
PC7	DHCP (client, via DHK)	DHCP	DHCP

4.4 Sylhet Board (SYL)

Device	IP Address	Subnet Mask	Default Gateway
Router SYL – Fa0/0 (central link)	10.79.6.3	255.255.255.248 (/29)	N/A
Router SYL – Gi6/0 (LAN)	10.79.4.1	255.255.255.0 (/24)	N/A
Server-PT EMAIL SYL	10.79.4.2	255.255.255.0	10.79.4.1
Printer1	10.79.4.10	255.255.255.0	10.79.4.1
PC0	DHCP (client, via DHK)	DHCP	DHCP
PC1	DHCP (client, via DHK)	DHCP	DHCP

4.5 Barishal Board (BAR)

Device	IP Address	Subnet Mask	Default Gateway
Router BAR – Fa0/0 (link to KHU)	10.79.6.18	255.255.255.252 (/30)	N/A
Router BAR – Fa1/0 (link to DHK)	10.79.6.22	255.255.255.252 (/30)	N/A
Router BAR – Gi6/0 (LAN)	10.79.5.1	255.255.255.128 (/25)	N/A
Server-PT DHCP BAR	10.79.5.2	255.255.255.128	10.79.5.1
Server-PT EMAIL BAR	10.79.5.4	255.255.255.128	10.79.5.1
Printer2	10.79.5.10	255.255.255.128	10.79.5.1
PC10	DHCP (client, via BAR server)	DHCP	DHCP
PC11	DHCP (client, via BAR server)	DHCP	DHCP

4.6 Khulna Board (KHU)

Device	IP Address	Subnet Mask	Default Gateway
Router KHU – Fa0/0 (link to DHK)	10.79.6.14	255.255.255.252 (/30)	N/A
Router KHU – Fa1/0 (link to BAR)	10.79.6.17	255.255.255.252 (/30)	N/A
Router KHU – Gi6/0 (LAN)	10.79.5.129	255.255.255.128 (/25)	N/A
Server-PT DHCP KHU	10.79.5.130	255.255.255.128	10.79.5.129
Server-PT EMAIL KHU	10.79.5.132	255.255.255.128	10.79.5.129
Printer0	10.79.5.250	255.255.255.128	10.79.5.129
PC2	DHCP (client, via KHU server)	DHCP	DHCP
PC3	DHCP (client, via KHU server)	DHCP	DHCP

5. Router Configuration Commands

The configurations below are the verbatim running-configuration of each router, extracted directly from the submitted Packet Tracer file (421_project_version_2.pkt).

5.1 Router DHK

Core router: main DHCP server for DHK/CTG/RAJ/SYL, RIPv2 with redistribute static, static routes for CTG and the floating/backup routes to KHU and BAR.

```
!  
version 12.2  
no service timestamps log datetime msec  
no service timestamps debug datetime msec  
no service password-encryption  
!  
hostname DHK  
!  
ip dhcp excluded-address 10.79.0.1 10.79.0.10  
ip dhcp excluded-address 10.79.2.1 10.79.2.10  
ip dhcp excluded-address 10.79.3.1 10.79.3.10  
ip dhcp excluded-address 10.79.4.1 10.79.4.10  
!  
ip dhcp pool DHK_POOL  
network 10.79.0.0 255.255.254.0  
default-router 10.79.0.1  
dns-server 10.79.0.2  
ip dhcp pool CTG_POOL  
network 10.79.2.0 255.255.255.0  
default-router 10.79.2.1  
dns-server 10.79.0.2  
ip dhcp pool RAJ_POOL  
network 10.79.3.0 255.255.255.0  
default-router 10.79.3.1  
dns-server 10.79.0.2  
ip dhcp pool SYL_POOL  
network 10.79.4.0 255.255.255.0  
default-router 10.79.4.1  
dns-server 10.79.0.2  
!  
no ip cef  
no ipv6 cef  
!  
interface FastEthernet0/0  
ip address 10.79.6.1 255.255.255.248  
duplex auto  
speed auto  
!  
interface FastEthernet1/0  
ip address 10.79.6.9 255.255.255.252  
duplex auto  
speed auto  
!  
interface Serial2/0  
no ip address  
clock rate 2000000  
shutdown  
!  
interface Serial3/0  
no ip address  
clock rate 2000000  
shutdown  
!  
interface FastEthernet4/0  
no ip address  
shutdown  
!  
interface FastEthernet5/0  
no ip address
```

```

shutdown
!
interface GigabitEthernet6/0
ip address 10.79.0.1 255.255.254.0
duplex auto
speed auto
!
interface GigabitEthernet7/0
ip address 10.79.6.13 255.255.255.252
duplex auto
speed auto
!
interface GigabitEthernet8/0
ip address 10.79.6.21 255.255.255.252
duplex auto
speed auto
!
router rip
version 2
redistribute static
network 10.0.0.0
no auto-summary
!
ip classless
ip route 10.79.2.0 255.255.255.0 10.79.6.10
ip route 10.79.5.128 255.255.255.128 10.79.6.14
ip route 10.79.5.128 255.255.255.128 10.79.6.22 10
ip route 10.79.5.0 255.255.255.128 10.79.6.22
ip route 10.79.5.0 255.255.255.128 10.79.6.14 10
!
ip flow-export version 9
!
line con 0
!
line aux 0
!
line vty 0 4
login
!
end

```

5.2 Router RAJ

RIPv2 peer with DHK and SYL over the shared central-switch segment; ip helper-address relays DHCP requests to DHK.

```
!  
version 12.2  
no service timestamps log datetime msec  
no service timestamps debug datetime msec  
no service password-encryption  
!  
hostname RAJ  
!  
no ip cef  
no ipv6 cef  
!  
interface FastEthernet0/0  
ip address 10.79.6.2 255.255.255.248  
duplex auto  
speed auto  
!  
interface FastEthernet1/0  
no ip address  
duplex auto  
speed auto  
shutdown  
!  
interface Serial2/0  
no ip address  
clock rate 2000000  
shutdown  
!  
interface Serial3/0  
no ip address  
clock rate 2000000  
shutdown  
!  
interface FastEthernet4/0  
no ip address  
shutdown  
!  
interface FastEthernet5/0  
no ip address  
shutdown  
!  
interface GigabitEthernet6/0  
ip address 10.79.3.1 255.255.255.0  
ip helper-address 10.79.0.1  
duplex auto  
speed auto  
!  
router rip  
version 2  
network 10.0.0.0  
no auto-summary  
!  
ip classless  
!  
ip flow-export version 9  
!  
line con 0  
!  
line aux 0  
!  
line vty 0 4  
login  
!  
end
```

5.3 Router SYL

RIPv2 peer with DHK and RAJ; ip helper-address relays DHCP requests to DHK.

```
!  
version 12.2  
no service timestamps log datetime msec  
no service timestamps debug datetime msec  
no service password-encryption  
!  
hostname SYL  
!  
ip cef  
no ipv6 cef  
!  
interface FastEthernet0/0  
 ip address 10.79.6.3 255.255.255.248  
 duplex auto  
 speed auto  
!  
interface FastEthernet1/0  
 no ip address  
 duplex auto  
 speed auto  
 shutdown  
!  
interface Serial2/0  
 no ip address  
 clock rate 2000000  
 shutdown  
!  
interface Serial3/0  
 no ip address  
 clock rate 2000000  
 shutdown  
!  
interface FastEthernet4/0  
 no ip address  
 shutdown  
!  
interface FastEthernet5/0  
 no ip address  
 shutdown  
!  
interface GigabitEthernet6/0  
 ip address 10.79.4.1 255.255.255.0  
 ip helper-address 10.79.0.1  
 duplex auto  
 speed auto  
!  
router rip  
 version 2  
 network 10.0.0.0  
 no auto-summary  
!  
ip classless  
!  
ip flow-export version 9  
!  
line con 0  
!  
line aux 0  
!  
line vty 0 4  
 login  
!  
end
```

5.4 Router CTG

Stub router — single default route to DHK (0.0.0.0/0 via 10.79.6.9), matching the "isolated board" requirement; DHCP requests relayed to DHK via ip helper-address.

```
!  
version 12.2  
no service timestamps log datetime msec  
no service timestamps debug datetime msec  
no service password-encryption  
!  
hostname CTG  
!  
no ip cef  
no ipv6 cef  
!  
interface FastEthernet0/0  
 ip address 10.79.6.10 255.255.255.252  
 duplex auto  
 speed auto  
!  
interface FastEthernet1/0  
 no ip address  
 duplex auto  
 speed auto  
 shutdown  
!  
interface Serial2/0  
 no ip address  
 clock rate 2000000  
 shutdown  
!  
interface Serial3/0  
 no ip address  
 clock rate 2000000  
 shutdown  
!  
interface FastEthernet4/0  
 no ip address  
 shutdown  
!  
interface FastEthernet5/0  
 no ip address  
 shutdown  
!  
interface GigabitEthernet6/0  
 ip address 10.79.2.1 255.255.255.0  
 ip helper-address 10.79.0.1  
 duplex auto  
 speed auto  
!  
ip classless  
ip route 0.0.0.0 0.0.0.0 10.79.6.9  
!  
ip flow-export version 9  
!  
line con 0  
!  
line aux 0  
!  
line vty 0 4  
 login  
!  
end
```

5.5 Router KHU

Static routing to all other networks via a default route to DHK, plus a specific floating static route to BAR's LAN (AD 10) for redundancy; default route to BAR (AD 10) is the automatic failover path if the DHK link fails.

```
!  
version 12.2  
no service timestamps log datetime msec  
no service timestamps debug datetime msec  
no service password-encryption  
!  
hostname KHU  
!  
no ip cef  
no ipv6 cef  
!  
interface FastEthernet0/0  
 ip address 10.79.6.14 255.255.255.252  
 duplex auto  
 speed auto  
!  
interface FastEthernet1/0  
 ip address 10.79.6.17 255.255.255.252  
 duplex auto  
 speed auto  
!  
interface Serial2/0  
 no ip address  
 clock rate 2000000  
 shutdown  
!  
interface Serial3/0  
 no ip address  
 clock rate 2000000  
 shutdown  
!  
interface FastEthernet4/0  
 no ip address  
 shutdown  
!  
interface FastEthernet5/0  
 no ip address  
 shutdown  
!  
interface GigabitEthernet6/0  
 ip address 10.79.5.129 255.255.255.128  
 duplex auto  
 speed auto  
!  
ip classless  
ip route 10.79.5.0 255.255.255.128 10.79.6.18  
ip route 10.79.5.0 255.255.255.128 10.79.6.13 10  
ip route 0.0.0.0 0.0.0.0 10.79.6.13  
ip route 0.0.0.0 0.0.0.0 10.79.6.18 10  
!  
ip flow-export version 9  
!  
line con 0  
!  
line aux 0  
!  
line vty 0 4  
 login  
!  
end
```

5.6 Router BAR

Static routing to all other networks via a default route to DHK, plus a specific floating static route to KHU's LAN (AD 10) for redundancy; default route to KHU (AD 10) is the automatic failover path if the DHK link fails.

```
!  
version 12.2  
no service timestamps log datetime msec  
no service timestamps debug datetime msec  
no service password-encryption  
!  
hostname BAR  
!  
no ip cef  
no ipv6 cef  
!  
interface FastEthernet0/0  
 ip address 10.79.6.18 255.255.255.252  
 duplex auto  
 speed auto  
!  
interface FastEthernet1/0  
 ip address 10.79.6.22 255.255.255.252  
 duplex auto  
 speed auto  
!  
interface Serial2/0  
 no ip address  
 clock rate 2000000  
 shutdown  
!  
interface Serial3/0  
 no ip address  
 clock rate 2000000  
 shutdown  
!  
interface FastEthernet4/0  
 no ip address  
 shutdown  
!  
interface FastEthernet5/0  
 no ip address  
 shutdown  
!  
interface GigabitEthernet6/0  
 ip address 10.79.5.1 255.255.255.128  
 duplex auto  
 speed auto  
!  
ip classless  
ip route 10.79.5.128 255.255.255.128 10.79.6.17  
ip route 10.79.5.128 255.255.255.128 10.79.6.21 10  
ip route 0.0.0.0 0.0.0.0 10.79.6.21  
ip route 0.0.0.0 0.0.0.0 10.79.6.17 10  
!  
ip flow-export version 9  
!  
line con 0  
!  
line aux 0  
!  
line vty 0 4  
 login  
!  
end
```

6. DNS Configuration

The Central DNS Server (Server-PT DNS DHK, 10.79.0.2) is enabled and holds the following A-records, verified directly from the .pkt file:

Record Type	Hostname	TTL	IP Address
A	www.dhk.edu.bd	86400	10.79.0.3
A	www.ctg.edu.bd	86400	10.79.2.2
A	mail.dhk.edu.bd	86400	10.79.0.4
A	mail.ctg.edu.bd	86400	10.79.2.3
A	mail.raj.edu.bd	86400	10.79.3.2
A	mail.syl.edu.bd	86400	10.79.4.2
A	mail.bar.edu.bd	86400	10.79.5.4
A	mail.khu.edu.bd	86400	10.79.5.132

All DHCP pools (main DHK router and the dedicated KHU/BAR servers) hand out 10.79.0.2 as the DNS server, so every board — including CTG, whose devices get addresses relayed through DHK — can resolve both domains and reach either web server by name.

7. DHCP Configuration

7.1 Main DHCP (DHK Router) — serves DHK, CTG, RAJ, SYL

```
ip dhcp excluded-address 10.79.0.1 10.79.0.10
ip dhcp excluded-address 10.79.2.1 10.79.2.10
ip dhcp excluded-address 10.79.3.1 10.79.3.10
ip dhcp excluded-address 10.79.4.1 10.79.4.10
```

```
ip dhcp pool DHK_POOL
 network 10.79.0.0 255.255.254.0
 default-router 10.79.0.1
 dns-server 10.79.0.2
ip dhcp pool CTG_POOL
 network 10.79.2.0 255.255.255.0
 default-router 10.79.2.1
 dns-server 10.79.0.2
ip dhcp pool RAJ_POOL
 network 10.79.3.0 255.255.255.0
 default-router 10.79.3.1
 dns-server 10.79.0.2
ip dhcp pool SYL_POOL
 network 10.79.4.0 255.255.255.0
 default-router 10.79.4.1
 dns-server 10.79.0.2
```

CTG's DHCP requests reach the DHK router because "ip helper-address 10.79.0.1" is configured on the CTG, RAJ, and SYL LAN interfaces, forwarding DHCP broadcasts across the WAN links to DHK's pools — satisfying the requirement that CTG devices obtain addresses through DHK even though CTG has no local DHCP service.

7.2 Dedicated DHCP — KHU Server (10.79.5.130)

Setting	Value
Network / Mask	10.79.5.128 / 255.255.255.128
Default Router	10.79.5.129
DNS Server	10.79.0.2
Address Pool	10.79.5.140 – 10.79.5.255
Status	Enabled — active leases confirmed in file (PC2, PC3)

7.3 Dedicated DHCP — BAR Server (10.79.5.2)

Setting	Value
Network / Mask	10.79.5.0 / 255.255.255.128
Default Router	10.79.5.1
DNS Server	10.79.0.2
Address Pool	10.79.5.10 – 10.79.5.127
Status	Enabled — active leases confirmed in file (PC10, PC11)

8. Web Server Configuration

Server	IP Address	HTTP Service	Domain (via DNS)
Server-PT WEB DHK	10.79.0.3	Enabled	www.dhk.edu.bd
Server-PT WEB CTG	10.79.2.2	Enabled	www.ctg.edu.bd

Both HTTP services are confirmed enabled in the device configuration. Because the DNS A-records for both domains are hosted centrally at DHK (10.79.0.2) and every board's devices are configured to use 10.79.0.2 as their DNS server, all boards — including CTG behind its single default route — can browse to either site by domain name.

9. Email Server Configuration

Every board runs its own local Email Server with SMTP and POP3 enabled, using the hostname format mail.<board>.edu.bd as required. Each server carries two verified accounts (admin / student) for cross-board testing.

Board	Server IP	SMTP Domain	SMTP	POP3	Accounts
DHK	10.79.0.4	mail.dhk.edu.bd	Enabled	Enabled	admin, student
CTG	10.79.2.3	mail.ctg.edu.bd	Enabled	Enabled	admin, student
RAJ	10.79.3.2	mail.raj.edu.bd	Enabled	Enabled	admin, student
SYL	10.79.4.2	mail.syl.edu.bd	Enabled	Enabled	admin, student
BAR	10.79.5.4	mail.bar.edu.bd	Enabled	Enabled	admin, student
KHU	10.79.5.132	mail.khu.edu.bd	Enabled	Enabled	admin, student

10. Testing & Verification Checklist

Requirement	Status (per .pkt inspection)
Topology matches spec (isolated CTG, DHK-RAJ-SYL central switch, DHK-KHU-BAR triangle)	Confirmed — matches exactly
Each board has 1 switch, 1 printer, 1 email server, exactly 2 PCs	Confirmed on all six boards
VLSM addressing, no overlaps	Confirmed — matches Section 3
DHK router configs (DHCP, RIP, static routes)	Confirmed present and correctly written
RAJ / SYL RIPv2 peering with DHK	Confirmed
CTG default route to DHK	Confirmed
KHU / BAR static + floating recursive redundancy routes	Confirmed, correctly implemented both directions
Central DNS (DHK) resolves both websites + all mail domains	Confirmed — 8 A-records present
Dedicated DHCP on KHU and BAR	Confirmed enabled, with live leases
Web servers (DHK, CTG) HTTP enabled	Confirmed
Email servers on all 6 boards, correct hostnames	Confirmed
All PCs receive an address (static or DHCP)	Fixed and confirmed — PC4 now set to DHCP, matching all other PCs
Every device can ping every other device	Re-verify live in Packet Tracer simulation mode before the demo
Redundancy demo (shut DHK-KHU / DHK-BAR links)	Routes are correctly configured for this; must be demonstrated live in Packet Tracer simulation mode